

DAVID KURNIAWAN

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EDUCATION

Xiamen University Malaysia | Selangor, Malaysia

Sept 2028

BEng (Hons) in Artificial Intelligence (GPA: 3.83 out of 4.00)

Dean's List Awardee for 3 consecutive semesters — College of Artificial Intelligence and Robotics • Top 16% of cohort

TECHNICAL SKILLS

Languages: Python • HTML & CSS • JavaScript • C Programming • C++

ML / AI Frameworks: TensorFlow • Keras • Scikit-learn • Pandas • LangChain • OpenCV • PyTorch

Infra and Frameworks: FastAPI • REST API • ML-Agents • Docker • WSL • Flask • Node • React

Tools and Workflow: Git • Cursor • Antigravity • Claude Code • VS Code • PyCharm • Anaconda

Languages (Human): Indonesian (Native) • English (IELTS Band 7) • Chinese (Elementary)

SELECTED PROJECTS

Orion | FastAPI, LangGraph, LangSmith, GitHub Actions, Python

2026

Tech Lead / Backend (4-person team)

- Built a **multi-agent LLM workflow** using **LangGraph** and **FastAPI** to automate end-to-end expense claim processing, employees submit claims in natural language; six-stage agentic pipeline extracts, validates, and auto-approves claims reducing processing time by **~80%**.
- Designed and implemented the backend API using **FastAPI** with **Pydantic v2** for type-safe request/response contracts; integrated **LangSmith** observability for real-time workflow tracing and debugging across all six agent stages.
- Established comprehensive CI/CD pipeline (**GitHub Actions**) with dual test strategies: PR gate runs **unit + stub-integration tests with 85% coverage** (120 passing tests), and nightly cron job runs **live regression tests** against production APIs (**GLM-5.1**) with automatic secret rotation.
- Built deterministic policy engine and ledger tools (**Python, rapidfuzz, pypdf**) to handle document parsing (PDF/DOCX), duplicate detection via fuzzy matching, and persistent JSON-based claim records; collaborated with QA lead to establish test-driven architecture achieving **85% code coverage** (up from 80%).

NeuralVoid | Node.js, FastAPI, Scikit-Learn, Railway

2026

- Built an end-to-end TikTok behavioural analytics platform with a **25-feature ML pipeline** (session detection, binge streak, doomscroll velocity) using a soft-voting ensemble (**Logistic Regression, Random Forest, XGBoost**) achieving **~96%** accuracy, integrated with **Google Gemini AI** for clinical report generation via **FastAPI** backend
- Developed a responsive **React SPA** dashboard with interactive data visualisations (heatmaps, radar charts, area charts) deployed on **Vercel** with **Railway-hosted REST API**.

Handwritten Digit Recognition | TensorFlow, Keras

2024

- Designed and trained a CNN using **Keras** and **TensorFlow** on the MNIST dataset, achieving 98.5% test set accuracy, exceeding the standard benchmark.
- Applied full data preprocessing, augmentation, and hyperparameter tuning pipeline to optimise model generalisation and convergence.

AWARDS AND CERTIFICATIONS

3rd Place — DPickleball AI Competitions (Unity • ML-Agents • Reinforcement Learning)

Oct 2025

XMUM AI Club • Xiamen University Malaysia

- Served as lead developer of 3-member team, architected the full training system design including environment setup via Gymnasium, custom reward shaping strategy, PPO algorithm implementation, and end-to-end training pipeline, directing technical execution across the competition team.
- Engineered the reward shaping function and PPO (Proximal Policy Optimisation) training loop using Unity ML-Agents and Gymnasium, iteratively tuning hyperparameters to achieve stable agent convergence and competitive autonomous play in digital pickleball environment.

Top 20% Globally — International Quant Championship, Stage 1

Apr 2025

World Quant BRAIN

- Placed top 20% globally in an international quantitative reasoning competition, benchmarking analytical and data-driven problem-solving skills against international competitors.

RESEARCH INTEREST

Artificial Intelligence • Agentic AI • Deep Learning • Quantitative Modelling • Robotic Learning • Sim-to-Real Reinforcement Learning • Autonomous Systems

LEADERSHIP & ORGANISATION EXPERIENCE

Student Council — Tritunggal Christian High School

Oct 2021 - Oct 2023

Head of Treasury Department

- Managed and reported school event finances across 10+ annual events including class meetings, Teacher's Day, Independence Day, and orientation, maintaining full financial accountability.
- Led a fundraising campaign through canteen sales initiatives, generating 4 million rupiah in revenue, demonstrating budget planning and cross-team coordination skills.